

# Question ID ee846db7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: ee846db7

A store sells two different-sized containers of a certain Greek yogurt. The store’s sales of this Greek yogurt totaled **1,277.94** dollars last month. The equation  **$5.48x + 7.30y = 1,277.94$**  represents this situation, where  **$x$**  is the number of smaller containers sold and  **$y$**  is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

- A. **5.48**
- B.  **$7.30y$**
- C. **7.30**
- D.  **$5.48x$**

# Question ID 5b8a8475

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 5b8a8475

Line  $k$  is defined by  $y = 3x + 15$ . Line  $j$  is perpendicular to line  $k$  in the  $xy$ -plane. What is the slope of line  $j$ ?

- A.  $-\frac{1}{3}$
- B.  $-\frac{1}{12}$
- C.  $-\frac{1}{18}$
- D.  $-\frac{1}{45}$

# Question ID b52e5b6f

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b52e5b6f

A mixture consisting of only vitamin D and calcium has a total mass of **150** grams. The mass of vitamin D in the mixture is **50** grams. What is the mass, in grams, of calcium in the mixture?

- A. **200**
- B. **150**
- C. **100**
- D. **50**

# Question ID 83f2c3bf

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 83f2c3bf

$$y = x + 4$$

Which table gives three values of  $x$  and their corresponding values of  $y$  for the given equation?

A.

| $x$ | $y$ |
|-----|-----|
| 0   | 4   |
| 1   | 5   |
| 2   | 6   |

B.

| $x$ | $y$ |
|-----|-----|
| 0   | 6   |
| 1   | 5   |
| 2   | 4   |

C.

| $x$ | $y$ |
|-----|-----|
| 0   | 2   |
| 1   | 1   |
| 2   | 0   |

D.

| $x$ | $y$ |
|-----|-----|
| 0   | 0   |
| 1   | 1   |
| 2   | 2   |

# Question ID 4d8ccb96

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 4d8ccb96

A chemist studying the impact of salt on a process mixes  $x$  kilograms of a low-salt mixture, which is **2%** salt by weight, with  $y$  kilograms of a high-salt mixture, which is **96%** salt by weight, to create **24** kilograms of a mixture that is **4%** salt by weight. Which equation represents this situation?

- A.  $0.96x + 0.02y = (0.04)(24)$
- B.  $0.02x + 0.96y = (0.04)(24)$
- C.  $0.96x + 0.02y = 24$
- D.  $0.02x + 0.96y = 24$

# Question ID b23bba4c

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b23bba4c

$3a + 4b = 25$

A shipping company charged a customer \$25 to ship some small boxes and some large boxes. The equation above represents the relationship between  $a$ , the number of small boxes, and  $b$ , the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

- A. 3
- B. 4
- C. 5
- D. 6

# Question ID 24854644

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 24854644

What is the equation of the line that passes through the point  $(0, 5)$  and is parallel to the graph of  $y = 7x + 4$  in the  $xy$ -plane?

- A.  $y = 5x$
- B.  $y = 7x + 5$
- C.  $y = 7x$
- D.  $y = 5x + 7$

# Question ID 87322577

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 87322577

$x + y = 75$

The equation above relates the number of minutes,  $x$ , Maria spends running each day and the number of minutes,  $y$ , she spends biking each day. In the equation, what does the number 75 represent?

- A. The number of minutes spent running each day
- B. The number of minutes spent biking each day
- C. The total number of minutes spent running and biking each day
- D. The number of minutes spent biking for each minute spent running



# Question ID ebf8d2b7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: ebf8d2b7

A machine makes large boxes or small boxes, one at a time, for a total of **700** minutes each day. It takes the machine **10** minutes to make a large box or **5** minutes to make a small box. Which equation represents the possible number of large boxes, *x*, and small boxes, *y*, the machine can make each day?

- A.  $5x + 10y = 700$
- B.  $10x + 5y = 700$
- C.  $(x + y)(10 + 5) = 700$
- D.  $(10 + x)(5 + y) = 700$

# Question ID 6fa1dc0f

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 6fa1dc0f

Line  $r$  in the  $xy$ -plane has a slope of  $4$  and passes through the point  $(0, 6)$ . Which equation defines line  $r$ ?

- A.  $y = -6x + 4$
- B.  $y = 6x + 4$
- C.  $y = 4x - 6$
- D.  $y = 4x + 6$

# Question ID c6b151d4

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c6b151d4

A total of **364** paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation  $3x + 4y = 364$  represents this situation, where  $x$  is the number of triangles constructed and  $y$  is the number of rectangles constructed. What is the best interpretation of  $(x, y) = (24, 73)$  in this context?

- A. If **24** triangles were constructed, then **73** rectangles were constructed.
- B. If **24** triangles were constructed, then **73** paper straws were used.
- C. If **73** triangles were constructed, then **24** rectangles were constructed.
- D. If **73** triangles were constructed, then **24** paper straws were used.

# Question ID 8c98c834

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 8c98c834

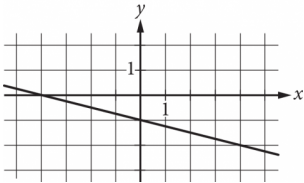
The equation  $y = 0.1x$  models the relationship between the number of different pieces of music a certain pianist practices,  $y$ , during an  $x$ -minute practice session. How many pieces did the pianist practice if the session lasted 30 minutes?

- A. 1
- B. 3
- C. 10
- D. 30

Question ID b2845d88

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b2845d88



Which of the following is an equation of the graph shown in the  $xy$ -plane above?

- A.  $y = -\frac{1}{4}x - 1$
- B.  $y = -x - 4$
- C.  $y = -x - \frac{1}{4}$
- D.  $y = -4x - 1$

# Question ID b450ab03

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b450ab03

An employee at a restaurant prepares sandwiches and salads. It takes the employee **1.5** minutes to prepare a sandwich and **1.9** minutes to prepare a salad. The employee spends a total of **46.1** minutes preparing *x* sandwiches and *y* salads. Which equation represents this situation?

- A.  $1.9x + 1.5y = 46.1$
- B.  $1.5x + 1.9y = 46.1$
- C.  $x + y = 46.1$
- D.  $30.7x + 24.3y = 46.1$

# Question ID db0107df

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

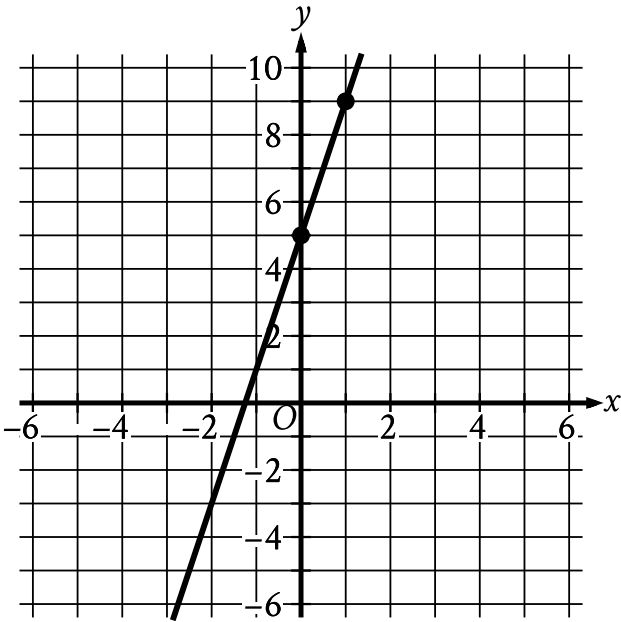
ID: db0107df

The y-intercept of the graph of  $12x + 2y = 18$  in the  $xy$ -plane is  $(0, y)$ . What is the value of  $y$ ?

Question ID b8fa27db

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b8fa27db



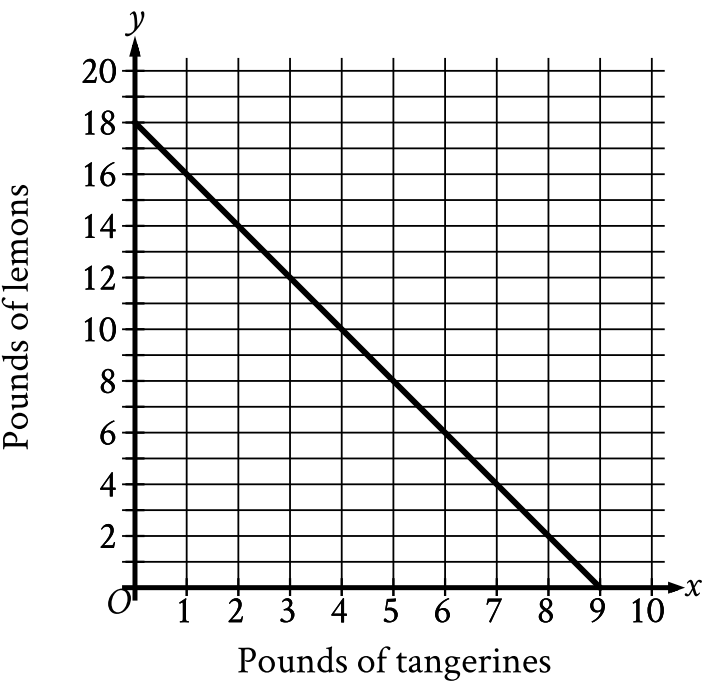
Line  $j$  is shown in the  $xy$ -plane. Line  $k$  (not shown) is parallel to line  $j$ . What is the slope of line  $k$ ?



Question ID 8368afd1

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 8368afd1



The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for **\$18** at a certain store. If Melvin purchased lemons and **4** pounds of tangerines for a total of **\$18**, how many pounds of lemons did he purchase?

- A. **7**
- B. **10**
- C. **14**
- D. **16**

# Question ID 8adf1335

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 8adf1335

A city’s total expense budget for one year was  $x$  million dollars. The city budgeted  $y$  million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between  $x$  and  $y$  in this context?

- A.  $x + y = 201$
- B.  $x - y = 201$
- C.  $2x - y = 201$
- D.  $y - x = 201$

Question ID e7343559

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: e7343559

$y = -4x + 40$

Which table gives three values of  $x$  and their corresponding values of  $y$  for the given equation?

A.

| $x$ | $y$ |
|-----|-----|
| 0   | 0   |
| 1   | -4  |
| 2   | -8  |

B.

| $x$ | $y$ |
|-----|-----|
| 0   | 40  |
| 1   | 44  |
| 2   | 48  |

C.

| $x$ | $y$ |
|-----|-----|
| 0   | 40  |
| 1   | 36  |
| 2   | 32  |

D.

| $x$ | $y$ |
|-----|-----|
| 0   | 0   |
| 1   | 4   |
| 2   | 8   |

# Question ID d1042cf8

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: d1042cf8

A food truck buys forks for ~~\$0.04~~ each and plates for ~~\$0.48~~ each. The total cost of  $x$  forks and  $y$  plates is ~~\$661.76~~. Which equation represents this situation?

- A.  $0.48x - 0.04y = 661.76$
- B.  $0.04x - 0.48y = 661.76$
- C.  $0.48x + 0.04y = 661.76$
- D.  $0.04x + 0.48y = 661.76$

# Question ID dd797fe2

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: dd797fe2

$4x + 3y = 24$

Mario purchased 4 binders that cost  $x$  dollars each and 3 notebooks that cost  $y$  dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

- A. The total cost, in dollars, for all binders purchased
- B. The total cost, in dollars, for all notebooks purchased
- C. The total cost, in dollars, for all binders and notebooks purchased
- D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

# Question ID 80ae6851

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 80ae6851

A producer is creating a video with a length of **70** minutes. The video will consist of segments that are **1** minute long and segments that are **3** minutes long. Which equation represents this situation, where *x* represents the number of **1**-minute segments and *y* represents the number of **3**-minute segments?

- A.  $4xy = 70$
- B.  $4(x + y) = 70$
- C.  $3x + y = 70$
- D.  $x + 3y = 70$

# Question ID 789975b7

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 789975b7

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where  $x$  is the number of pounds of Fertilizer A and  $y$  is the number of pounds of Fertilizer B?

- A.  $0.4x + 0.6y = 240$
- B.  $0.6x + 0.4y = 240$
- C.  $40x + 60y = 240$
- D.  $60x + 40y = 240$

# Question ID cea27ab2

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: cea27ab2

$$7x - 4y = -84$$

For the given equation, which table gives three values of  $x$  and their corresponding values of  $y$ ?

- A.

|     |    |    |    |
|-----|----|----|----|
| $x$ | 0  | 4  | 8  |
| $y$ | 21 | 28 | 35 |
- B.

|     |    |    |    |
|-----|----|----|----|
| $x$ | 0  | 4  | 8  |
| $y$ | 35 | 28 | 21 |
- C.

|     |    |    |    |
|-----|----|----|----|
| $x$ | 21 | 28 | 35 |
| $y$ | 0  | 4  | 8  |
- D.

|     |    |    |    |
|-----|----|----|----|
| $x$ | 21 | 28 | 35 |
| $y$ | 8  | 4  | 0  |



# Question ID 2554b413

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 2554b413

In the  $xy$ -plane, a line has a slope of 6 and passes through the point  $(0,8)$ . Which of the following is an equation of this line?

- A.  $y = 6x + 8$
- B.  $y = 6x + 48$
- C.  $y = 8x + 6$
- D.  $y = 8x + 48$

# Question ID 52a8ef85

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 52a8ef85

The equation  $40x + 20y = 160$  represents the number of sweaters,  $x$ , and number of shirts,  $y$ , that Yesenia purchased for \$160. If Yesenia purchased 2 sweaters, how many shirts did she purchase?

- A. 3
- B. 4
- C. 8
- D. 40

# Question ID ba79f10f

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: ba79f10f

| $x$ | $y$ |
|-----|-----|
| 0   | 18  |
| 1   | 13  |
| 2   | 8   |

The table shows three values of  $x$  and their corresponding values of  $y$ . There is a linear relationship between  $x$  and  $y$ . Which of the following equations represents this relationship?

- A.  $y = 18x + 13$
- B.  $y = 18x + 18$
- C.  $y = -5x + 13$
- D.  $y = -5x + 18$

# Question ID dfa45424

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: dfa45424

Tony spends \$80 per month on public transportation. A 10-ride pass costs \$12.50, and a single-ride pass costs \$1.50. If  $g$  represents the number of 10-ride passes Tony buys in a month and  $t$  represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between  $g$  and  $t$  ?

- A.  $g + t = 80$
- B.  $g + t = 1.50 + 12.50$
- C.  $1.50g + 12.50t = 80$
- D.  $12.50g + 1.50t = 80$

# Question ID 6a12efbb

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 6a12efbb

The equation  $46 = 2x + 2y$  gives the perimeter of a rectangular rug that has length  $x$ , in feet, and width  $y$ , in feet. The width of the rug is 8 feet. What is the length, in feet, of the rug?

# Question ID 10c448d6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 10c448d6

A line in the  $xy$ -plane has a slope of  $\frac{1}{9}$  and passes through the point  $(0, 14)$ . Which equation represents this line?

- A.  $y = -\frac{1}{9}x - 14$
- B.  $y = -\frac{1}{9}x + 14$
- C.  $y = \frac{1}{9}x - 14$
- D.  $y = \frac{1}{9}x + 14$

# Question ID 7038b587

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

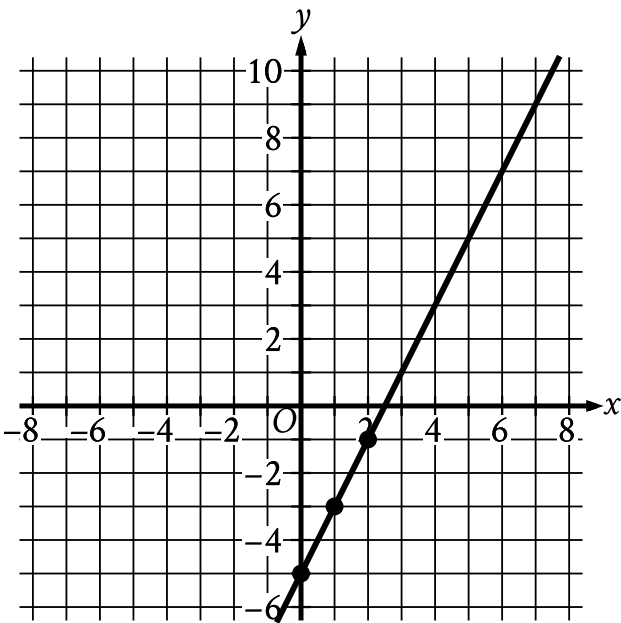
ID: 7038b587

Vivian bought party hats and cupcakes for **\$71**. Each package of party hats cost **\$3**, and each cupcake cost **\$1**. If Vivian bought **10** packages of party hats, how many cupcakes did she buy?

Question ID 4acd05cd

|            |      |         |                                   |                                              |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 4acd05cd



The graph shows the linear relationship between  $x$  and  $y$ . Which table gives three values of  $x$  and their corresponding values of  $y$  for this relationship?

A.

| $x$ | $y$ |
|-----|-----|
| 0   | 0   |
| 1   | -7  |
| 2   | -9  |

B.

| $x$ | $y$ |
|-----|-----|
| 0   | 0   |
| 1   | -3  |
| 2   | -1  |

C.

| $x$ | $y$ |
|-----|-----|
| 0   | -5  |
| 1   | -7  |
| 2   | -9  |



D.

| $x$ | $y$ |
|-----|-----|
| 0   | -5  |
| 1   | -3  |
| 2   | -1  |

Question ID b2de69bd

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b2de69bd

| x | y  |
|---|----|
| 1 | 5  |
| 2 | 7  |
| 3 | 9  |
| 4 | 11 |

The table above shows some pairs of  $x$  values and  $y$  values. Which of the following equations could represent the relationship between  $x$  and  $y$  ?

- A.  $y = 2x + 3$
- B.  $y = 3x - 2$
- C.  $y = 4x - 1$
- D.  $y = 5x$

Question ID 5518885d

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 5518885d

The equation  $x + y = 1,440$  represents the number of minutes of daylight (between sunrise and sunset),  $x$ , and the number of minutes of non-daylight,  $y$ , on a particular day in Oak Park, Illinois. If this day has **670** minutes of daylight, how many minutes of non-daylight does it have?

- A. **670**
- B. **770**
- C. **1,373**
- D. **1,440**

# Question ID 2d0e13a6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 2d0e13a6

Line  $k$  is defined by  $y = \frac{1}{4}x + 1$ . Line  $j$  is parallel to line  $k$  in the  $xy$ -plane. What is the slope of  $j$ ?

# Question ID c8e0f511

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c8e0f511

For a camping trip a group bought  $x$  one-liter bottles of water and  $y$  three-liter bottles of water, for a total of **240** liters of water. Which equation represents this situation?

- A.  $x + 3y = 240$
- B.  $x + y = 240$
- C.  $3x + 3y = 240$
- D.  $3x + y = 240$

# Question ID 029c2dc2

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 029c2dc2

A teacher is creating an assignment worth **70** points. The assignment will consist of questions worth **1** point and questions worth **3** points. Which equation represents this situation, where ***x*** represents the number of **1**-point questions and ***y*** represents the number of **3**-point questions?

- A.  **$4xy = 70$**
- B.  **$4(x + y) = 70$**
- C.  **$3x + y = 70$**
- D.  **$x + 3y = 70$**

# Question ID c5479c1a

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: c5479c1a

A shipment consists of **5**-pound boxes and **10**-pound boxes with a total weight of **220** pounds. There are **13** **10**-pound boxes in the shipment. How many **5**-pound boxes are in the shipment?

- A. **5**
- B. **10**
- C. **13**
- D. **18**

# Question ID 1efd8202

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 1efd8202

$$y = 70x + 8$$

Which table gives three values of  $x$  and their corresponding values of  $y$  for the given equation?

A.

| $x$ | $y$ |
|-----|-----|
| 0   | 8   |
| 2   | 148 |
| 4   | 288 |

B.

| $x$ | $y$ |
|-----|-----|
| 0   | 70  |
| 2   | 78  |
| 4   | 86  |

C.

| $x$ | $y$ |
|-----|-----|
| 0   | 70  |
| 2   | 140 |
| 4   | 280 |

D.

| $x$ | $y$ |
|-----|-----|
| 0   | 8   |
| 2   | 132 |
| 4   | 272 |



# Question ID ed856e9c

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: ed856e9c

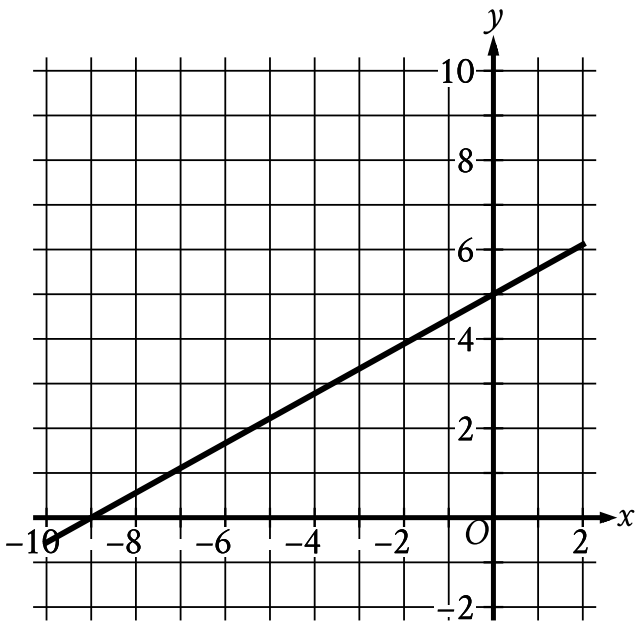
What is the y-intercept of the graph of  $y = 34x + 81$  in the xy-plane?

- A.  $(0, 81)$
- B.  $(0, 34)$
- C.  $(0, -34)$
- D.  $(0, -81)$

Question ID eeebe166

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: eeebe166



What is the y-intercept of the line graphed?

- A.  $(-5, 0)$
- B.  $(0, 0)$
- C.  $(0, 5)$
- D.  $(0, 9)$

# Question ID b9839f9e

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: b9839f9e

$F = 2.50x + 7.00y$

In the equation above,  $F$  represents the total amount of money, in dollars, a food truck charges for  $x$  drinks and  $y$  salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

- A. The price, in dollars, of one drink
- B. The price, in dollars, of one salad
- C. The number of drinks bought during the day
- D. The number of salads bought during the day

# Question ID 1fe778dc

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 1fe778dc

A line in the  $xy$ -plane has a slope of  $-\frac{1}{2}$  and passes through the point  $(0, 3)$ . Which equation represents this line?

- A.  $y = -\frac{1}{2}x - 3$
- B.  $y = -\frac{1}{2}x + 3$
- C.  $y = \frac{1}{2}x - 3$
- D.  $y = \frac{1}{2}x + 3$

Question ID 39617468

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 39617468

$$x + y = 350$$

The given equation relates the total number of maple trees,  $x$ , and the total number of birch trees,  $y$ , planted in a ~~14~~-acre forest preserve. If ~~24~~5 maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

- A. ~~14~~
- B. ~~25~~
- C. ~~105~~
- D. ~~245~~

# Question ID 174885f8

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 174885f8

Jay walks at a speed of **3** miles per hour and runs at a speed of **5** miles per hour. He walks for *w* hours and runs for *r* hours for a combined total of **14** miles. Which equation represents this situation?

- A.  $3w + 5r = 14$
- B.  $\frac{1}{3}w + \frac{1}{5}r = 14$
- C.  $\frac{1}{3}w + \frac{1}{5}r = 112$
- D.  $3w + 5r = 112$

# Question ID 8da536c6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 8da536c6

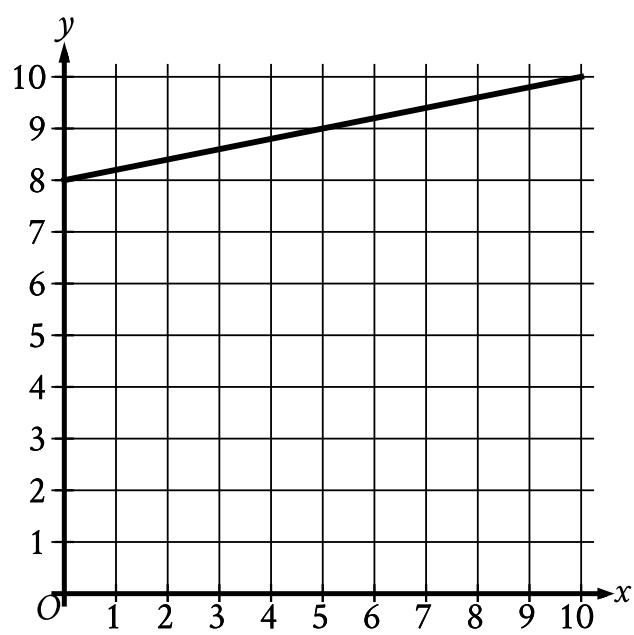
In **2010**, a swim club had a total of **35** swimmers, each classified as either advanced or intermediate. From **2010** to **2020**, the number of advanced swimmers in the club increased by approximately **53%**, and the number of intermediate swimmers in the club increased by approximately **44%**. The total number of swimmers in the club increased by approximately **49%**. Which equation best represents this situation, where *a* represents the number of advanced swimmers in the club in **2010** and *b* represents the number of intermediate swimmers in the club in **2010**?

- A.  $1.53a + 1.49b = 35(1.44)$
- B.  $1.49a + 0.53b = 35(1.44)$
- C.  $1.53a + 1.44b = 35(1.49)$
- D.  $1.44a + 1.53b = 35(1.49)$

Question ID f40552a9

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: f40552a9



What is the y-intercept of the line graphed?

- A.  $(0, -8)$
- B.  $(0, -\frac{1}{8})$
- C.  $(0, 0)$
- D.  $(0, 8)$



# Question ID 12ae3452

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 12ae3452

The equation  $46 = 2a + 2b$  gives the relationship between the side lengths  $a$  and  $b$  of a certain parallelogram. If  $a = 9$ , what is the value of  $b$ ?

# Question ID 39571c77

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 39571c77

Naomi bought both rabbit snails and nerite snails for a total of **\$52**. Each rabbit snail costs **\$8** and each nerite snail costs **\$6**. If Naomi bought **2** nerite snails, how many rabbit snails did she buy?

- A. 5
- B. 12
- C. 14
- D. 50

# Question ID 768b2425

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 768b2425

Last week, an interior designer earned a total of **\$1,258** from consulting for  $x$  hours and drawing up plans for  $y$  hours. The equation  $68x + 85y = 1,258$  represents this situation. Which of the following is the best interpretation of **68** in this context?

- A. The interior designer earned **\$68** per hour consulting last week.
- B. The interior designer worked **68** hours drawing up plans last week.
- C. The interior designer earned **\$68** per hour drawing up plans last week.
- D. The interior designer worked **68** hours consulting last week.

# Question ID 8b2a2a63

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

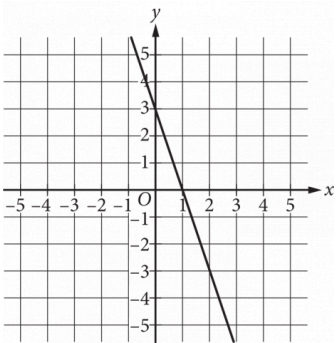
ID: 8b2a2a63

The y-intercept of the graph of  $y = -6x - 32$  in the xy-plane is  $(0, y)$ . What is the value of  $y$ ?

Question ID 8a1544f1

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 8a1544f1



What is the equation of the line shown in the  $xy$ -plane above?

- A.  $y = 3x - 3$
- B.  $y = -3x + 3$
- C.  $y = \frac{1}{3}x - 3$
- D.  $y = -\frac{1}{3}x + 3$

# Question ID 535fa6e6

| Assessment | Test | Domain  | Skill                             | Difficulty                                   |
|------------|------|---------|-----------------------------------|----------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | <div><div></div><div></div><div></div></div> |

ID: 535fa6e6

Davio bought some potatoes and celery. The potatoes cost **\$0.69** per pound, and the celery cost **\$0.99** per pound. If Davio spent **\$5.34** in total and bought twice as many pounds of celery as pounds of potatoes, how many pounds of celery did Davio buy?

- A. **2**
- B. **2.5**
- C. **2.67**
- D. **4**